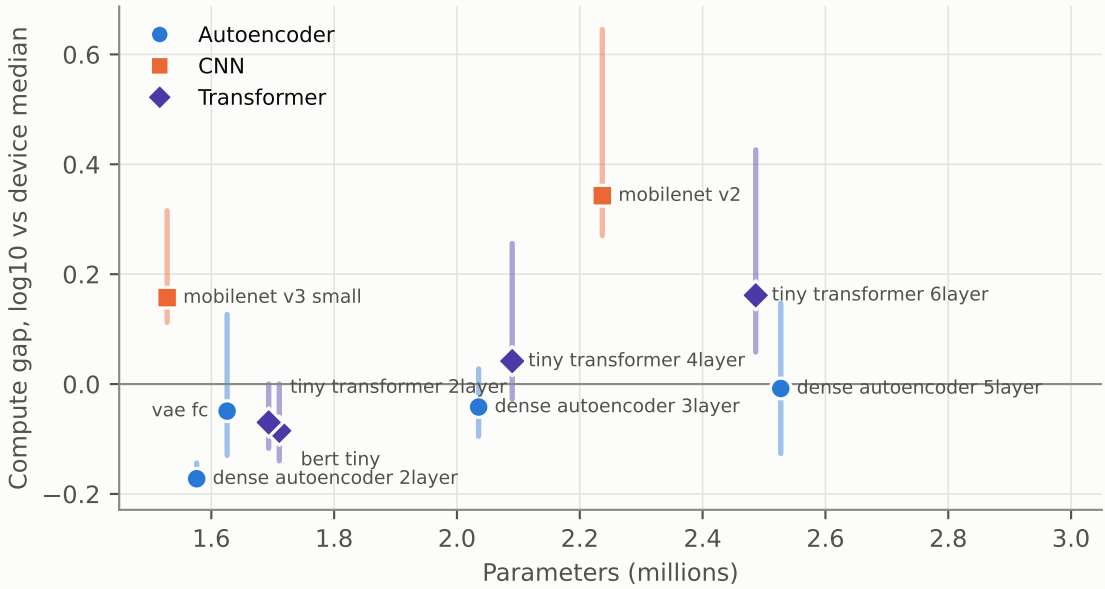
At matched size, CNNs show the longest compute gap
Autoencoder
CNN
Transformer
Compute gap, log10 vs device median
0.6
0.4
0.2
0.0
−0.2
mobilenet v2
mobilenet v3 small
tiny transformer 6layer
tiny transformer 4layer
tiny transformer 2layer
vae fc
dense autoencoder 3layer
dense autoencoder 5layer
bert tiny
dense autoencoder 2layer
1.6
1.8
2.0
2.2
2.4
2.6
2.8
3.0
Parameters (millions)